A Linear Programming Perspective on the Frankl–Rödl–Pippenger Theorem*

Jeff Kahn
Department of Mathematics and RUTCOR, Rutgers University, New Brunswick, NJ 08903

ABSTRACT

For $\mathscr{H}$ a hypergraph on vertex set V and $t:\mathscr{H}\rightarrow \mathbf{R}^{+}$, set

$$\alpha(t)=\max\left\{\sum \{t(A):x,\,y\in A\in \mathscr{H}\}:x,\,y\in V, x\neq y\right\}.$$

We give a simple proof, based on an argument of Pippenger and Spencer [19] of the following rather general statement.

Theorem 1.5. *Fix k and l. Let $\mathscr{H}$ be a k-bounded hypergraph and t a fractional matching of $\mathscr{H}$. Let, furthermore, $c_1,\ldots,c_l:\mathscr{H}\rightarrow \mathbf{R}^{+}$, each satisfying*

$$\sum_{A\in\mathscr{H}} c_i^2(A)t(A)<o\left(\left(\sum_{A\in\mathscr{H}} c_i(A)t(A)\right)^2\right).$$

Then there is matching $\mathscr{M}$ of $\mathscr{H}$ satisfying

$$\sum_{A\in\mathscr{M}} c_i(A)\sim c_i(\mathscr{H}) \qquad \text{for } i=1,\ldots,l,$$

limits taken as $\alpha(t)\rightarrow 0$.

This contains some previously announced results of the author which interpreted the theorem of the title as a statement about the relation between integer and linear programs and extended it in this vein. © 1996 John Wiley & Sons, Inc.

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1. LONG INTRODUCTION

The main purpose of this note is to give simple proofs of some fairly old results of the author (Theorems 1.2 and 1.4) which were stated in [10] and [16]. This is done in Section 2. We begin here with some context and motivation.

One of the most interesting and important combinatorial developments of recent years have been the discovery of methods for producing various, seemingly difficult to construct objects by means of small random or "semirandom" increments. A basic result in this vein, due to N. Pippenger (but see below for antecedents), is

Theorem 1.1. *Fix k. Let $\mathcal{H}$ be a k-uniform, D-regular hypergraph on vertex set V satisfying*

$$d(x, y) < o(D) \qquad \text{for all distinct } x, y \in V . \tag{1}$$

Then $\nu(\mathcal{H}) \sim n/k \sim \rho(\mathcal{H})$ as $D \to \infty$.

As usual ν and ρ denote matching and edge-cover numbers [so trivially $\nu(\mathcal{H}) \leq n/k \leq \rho(\mathcal{H})$]. We will always take V to be the vertex set of hypergraph $\mathcal{H}$, and $|V| = n$. For definitions see the end of this section.

Here and throughout we adopt the "uniformity convention" of [19]: Rates of convergence in our limiting statements depend on some specified numerical parameter or parameters, but not on the particular hypergraph, vertex, etc. in question. So, for example, Theorem 1.1 says that, with k fixed, there is, for each $\epsilon > 0$, a $\delta > 0$ so that, for any k-uniform, D-regular $\mathcal{H}$ with

$$d(x, y) < \delta D \qquad \text{for all distinct } x, y \in V ,$$

we have $\nu(\mathcal{H}) > (1 - \epsilon)n/k$, $\rho(\mathcal{H}) < (1 + \epsilon)n/k$.

Pippenger's theorem is a stronger and cleaner version of a theorem of Frankl and Rödl [5], itself a broad generalization of Rödl's proof [20] of the "Erdős–Hanani conjecture" [3]. Rödl's work in turn had roots in [1] and [18]. See [6], [10], [13], and [14] for further discussion of these and related topics, as well as [7] and [17] for more recent, major developments.

While Pippenger never published his original proof of Theorem 1.1, the argument may be found in [22] and [6]. In [19] Pippenger and Spencer give an elegant proof which shows more, in particular that a suitable random procedure usually produces a matching of the desired size. See Section 2 for details.

For $t : \mathcal{H} \to \mathbf{R}^+$, set $t(\mathcal{H}) = \Sigma \ \{t(A) : A \in \mathcal{H}\}$, and for $W \subseteq V$, $\bar{t}(W) = \Sigma \ \{t(A) : W \subseteq A \in \mathcal{H}\}$. Set

$$\alpha(t) = \max\{\bar{t}(\{x, y\}) : x, y \in V, x \neq y\} ,$$

a fractional analogue of the maximum of the codegrees $d(x, y)$. Recall that t is a *fractional matching* of $\mathcal{H}$ if

$$(\bar{t}(x) =) \qquad \sum_{A \ni x} t(A) \leq 1 \qquad \forall x \in V , \tag{2}$$

and a *fractional cover* if the inequalities in (2) are reversed.

About 6 years ago the author proved the following generalization of Theorem 1.1.

Theorem 1.2. *Fix k.*

(*a*) *If $\mathscr{H}$ is k-bounded and t is a fractional matching of $\mathscr{H}$, then*

$$\nu(\mathscr{H}) \gtrsim t(\mathscr{H}) \qquad (\alpha(t) \to 0) .$$

(*b*) *If instead t is a fractional cover of $\mathscr{H}$, then*

$$\rho(H) \lesssim t(\mathscr{H}) \qquad (\alpha(t) \to 0) .$$

Again note that convergence—e.g., of the lower bound on $\nu(\mathscr{H})/t(\mathscr{H})$ to 1—is uniform in $\mathscr{H}$. Of course, we get Theorem 1.1 from Theorem 1.2 by taking $t \equiv 1/D$.

The original motivation for Theorem 1.2 was its relevance to the following problem of Erdős and Lovász (see [4] or, e.g., [2]; the form given here is dual to that in these references).

For positive integer r, let

$$n(r) = \min\{|V| : \mathscr{H}\ r\text{-regular}; d_{\mathscr{H}}(x, y) \geq 1 \quad \forall x, y \in V ; \quad \rho(\mathscr{H}) = r\} .$$

Notice that the conditions

$$r\text{-regular} , \quad d_{\mathscr{H}}(x, y) \geq 1 \tag{3}$$

trivially give $\rho(\mathscr{H}) \leq r$, so $n(r)$ measures the difficulty of forcing ρ to be large. The basic question in [4] concerning $n(r)$ was: Does

$$n(r)/r \to \infty \qquad \text{as } r \to \infty? \tag{4}$$

Theorem 1.2 gives (4) provided we restrict to $\mathscr{H}$ with small codegrees:

Corollary 1.3. *Suppose $\mathscr{H}$ is r-regular with at most cr vertices, c fixed, $d(x, y) \geq 1$ for all $x, y \in V$, and*

$$\max\{d(x, y) : x, y \in V, x \neq y\} = o(r) .$$

Then $\rho(\mathscr{H}) < (c/(c+1) + o(1))r$.

The connection with Theorem 1.2(b) is the observation that, under the conditions (3), $t(A) := |A|/(n + r - 1)$ is a fractional cover with $t(\mathscr{H}) \sim nr/(n + r) \leq (c/(c + 1))r$. This gives Corollary 1.3 immediately when $\mathscr{H}$ is k-bounded (k fixed), though some additional argument is needed in general (see [10], Theorem 8.2).

These observations were originally thought to be tending to a proof of (4), particularly as the best hope for counterexamples seemed (in retrospect, naively) to involve $\mathscr{H}$'s with small $d(x, y)$'s; and indeed the best constructions [4, 9] were based on projective planes and had all codegrees equal to 1. As it turns out (see [11]), $n(r)$, rather surprisingly, grows only linearly in r; still, understanding Theorem 1.2 and Corollary 1.3 was crucial in arriving at the construction of [11].

The original proof of Theorem 1.2 was sketched in [10]. It is along the lines of Pippenger's original proof of Theorem 1.1 (actually, as noted in [10], the fractional viewpoint makes things a little easier for proofs along these lines), and the author apparently never felt sufficiently motivated to publish it: The contribution was thought to be mainly the point of view, i.e., that Theorem 1.1 should be regarded as a statement about fractional vs. integer behavior of certain linear programs. This viewpoint has indeed proved fruitful (see, e.g., [16] and [15]). In particular [16] proves a much more substantial (than Theorem 1.2) generalization of Theorem 1.1, the starting point of which is the following extension of Theorem 1.2(b).

Theorem 1.4. *Fix k. Let $\mathcal{H}$ be a k-bounded hypergraph and t a fractional matching of $\mathcal{H}$. Then there exists $\mathcal{A} \subseteq \mathcal{H}$ with*

$$|\mathcal{A}| \lesssim t(\mathcal{H}) \qquad \text{and} \qquad |\cup \mathcal{A}| \gtrsim \sum_{x \in V} \bar{t}(x)\,,$$

limits taken as $\alpha(t) \to 0$.

For the derivation of Theorem 1.2(b): It is easy to see (see [10, Proposition 8.3]) that in Theorem 1.2(b) we may assume that t is also a fractional *matching*. (That is, we have equality in (2); t is then a *fractional tiling*. Note the analogous reduction in Theorem 1.2(a) is *not* valid.) Then the $\mathcal{A}$ given by Theorem 1.4 covers all but at most $o(n)$ vertices, and so may be extended to a cover by addition of at most $o(n)$ edges (an acceptable amount since $t(\mathcal{H}) \geq n/k$).

Regrettably, the author has never published the proof of Theorem 1.4 either. Very recently Spencer [24] extended his beautiful "continuous" proof [23] of Theorem 1.1 to give a proof of Theorem 1.2 for uniform $\mathcal{H}$. In thinking over his proof, the author noticed that Theorem 1.2 actually follows easily from the Pippenger–Spencer proof of Theorem 1.1—though not, seemingly, from the statement—and that a little elaboration of the key point in that proof yields Theorem 1.4 and more:

Theorem 1.5. *Fix k and l. Let $\mathcal{H}$ be a k-bounded hypergraph and t a fractional matching of $\mathcal{H}$. Let, furthermore, $c_1, \ldots, c_l : \mathcal{H} \to \mathbf{R}^+$, each satisfying*

$$\sum_{A \in \mathcal{H}} c_i^2(A)t(A) < o\left(\left(\sum_{A \in \mathcal{H}} c_i(A)t(A)\right)^2\right). \tag{5}$$

Then there is matching $\mathcal{M}$ of $\mathcal{H}$ satisfying

$$\sum_{A \in \mathcal{M}} c_i(A) \sim \sum_{A \in \mathcal{H}} c_i(A)t(A) \qquad \text{for } i = 1, \ldots, l,$$

limits taken as $\alpha(t) \to 0$.

For example, (5) holds whenever $|c_i(A)|$ is bounded by some constant and Σ_A $c_i(A)t(A) \to \infty$. We recover Theorem 1.2(a) by taking ($l = 1$ and) $c_1 \equiv 1$, and Theorem 1.4 by taking $c_1 \equiv 1$ and $c_2(A) = |A|$.

(Strictly speaking, these derivations require the additional assumption Σ $t(A) \to \infty$ in Theorems 1.2, 1.4. But when $\Sigma\, t(A) = O(1)$, the results [actually with *exact* inequalities for small enough $\alpha(t)$ in place of the asymptotic inequalities] are easily established by elementary arguments, and are anyway not of much interest.

Also, as noted in Section 2, Theorem 1.2 in full is an easy case of our main argument.)

Theorem 1.5 is proved in Section 2.

Terminology

A *hypergraph* $\mathcal{H}$ is simply a collection of subsets of a finite set V. Elements of V are called *vertices* and elements of $\mathcal{H}$ *edges*. A hypergraph is *k-uniform* (*k-bounded*) if each of its edges has size k (at most k).

The *degree* in $\mathcal{H}$ of a vertex x is the number of edges containing x, and is denoted $d_{\mathcal{H}}(x)$ or simply $d(x)$. Similarly $d_{\mathcal{H}}(x, y)$ denotes the number of edges containing both of the vertices x, y. A hypergraph is *d-regular* if each of its vertices has degree d.

A *matching* is a collection of pairwise disjoint edges, and a *cover* is a collection of edges whose union is V. As mentioned above, we write $\nu(\mathcal{H})$ for the *matching number*, the maximum size of a matching in $\mathcal{H}$, and $\rho(\mathcal{H})$ for the cover number, the minimum size of a cover. For further hypergraph background see, for example, the excellent survey [6].

2. SHORT PROOF

The following lemma was stated without proof in [8] and [12]. Since it is in a way the main point of the present article (though less precise conditions on $\mathcal{F}$—e.g. approximate regularity, k'-uniformity for some fixed $k' \geq k$, small codegrees—would be enough for our purposes), we do reluctantly give a proof here.

Proposition 2.1. *For any k-bounded hypergraph $\mathcal{G}$ on V with degrees at most D, there is a k-uniform, D-regular hypergraph $\mathcal{F}$ on some $W \supseteq V$ satisfying*

(*a*) $\mathcal{F}|_V \supseteq \mathcal{G}$,

(*b*) *for all distinct* $x, y \in W$,

$$d_{\mathcal{F}}(x, y) \begin{cases} = d_{\mathcal{G}}(x, y) & \text{if } x, y \in V\,, \\ \leq 1 & \text{otherwise}\,. \end{cases}$$

Here $F|_V = \{A \cap V : A \in \mathcal{F}\}$, regarded as a *multiset* [so "$\supseteq$" in (a) also refers to multisets].

Proof. We construct $\mathcal{F}$ in several steps. At each step the new vertices and/or new edges introduced are all distinct.

Let $\mathcal{G}_1$ be obtained from $\mathcal{G}$ by adding $D - d_{\mathcal{G}}(x)$ copies of the singleton edge $\{x\}$ for each $x \in V$.

Let $\mathcal{G}_2$ be obtained from $\mathcal{G}_1$ by adding $k - |A|$ new points to each $A \in \mathcal{G}_1$. Notice that, with Y the set of vertices added in this step, $\mathcal{G}_2$ is k-uniform with degrees D in V and 1 in Y, so $k|\mathcal{G}_2| = D|V| + |Y|$ and

$$D|(k|\mathcal{G}_2| + (D-1)|Y|)\,. \tag{6}$$

Form $\mathscr{G}_3$ from $\mathscr{G}_2$ by adding

(i) a set M of m new vertices, m to be specified below, and
(ii) for each $x \in Y$ (resp. M) $D-1$ (resp. D) new edges, each consisting of x plus $k-1$ new vertices.

Let Z be the set of vertices added in (ii). Then $\mathscr{G}_3$ is k-uniform on $W := V \cup Y \cup M \cup Z$ with degrees D in $V \cup Y \cup M$ and 1 in Z. Moreover, the nonempty intersections of edges of $\mathscr{G}_3$ with Z form a partition of Z into sets, say $B_1, \ldots, B_t$, of size $k-1$.

To complete $\mathscr{G}_3$ to $\mathscr{F}$, we should add some k-uniform, $(D-1)$-regular $\mathscr{K}$ on Z satisfying

$$d_{\mathscr{K}}(x, y)\begin{cases} = 0 & \text{if } x, y \text{ are in the same } B_i\,, \\ \leq 1 & \text{otherwise}\,. \end{cases}$$

Note that since $t = (D-1)|Y| + Dm$ and [by (6)] $(k, D)|\ |Y|$, we may choose m as large as we like with t divisible by k.

Then, for example: Let $B_i = \{x_{i,1}, \ldots, x_{i,k-1}\}$ and $Z_j = \{x_{1,j}, \ldots, x_{t,j}\}$. Let $r = t/k$ and for $j \in [k-1]$ let $\mathscr{K}_j$ be a hypergraph on Z_j isomorphic to the hypergraph having vertex set the disjoint union, say $\mathbf{Z}_r^{(1)} \cup \cdots \cup \mathbf{Z}_r^{(k)}$, of k copies of $\mathbf{Z}_r$ and A an edge iff there are $x \in \mathbf{Z}_r$ and $a \in \{0, 1, \ldots, D-2\}$ such that A meets each $\mathbf{Z}_r^{(i)}$ in its copy of $x + (i-1)a$. [So we need $t \geq k(D-1)$.] Finally, set $\mathscr{K} = \cup \mathscr{K}_j$. □

To apply Proposition 2.1 in the situation of Theorems 1.2 and 1.5, we augment $\mathscr{H}$ as follows. We may of course assume the weights $t(A)$ are rational, say with common denominator D. Let $\mathscr{G}$ be the hypergraph on V consisting of $Dt(A)$ copies of each $A \in \mathscr{H}$. Then $\mathscr{G}$ is k-bounded with all degrees at most D and [as $\alpha(t) \to 0$]

$$d_{\mathscr{G}}(x, y) < o(D) \qquad \text{for distinct } x, y \in V\,.$$

We may thus extend $\mathscr{G}$ as in Proposition 2.1 to some $\mathscr{F}$ on $W \supseteq V$ which in particular satisfies (1) of Theorem 1.1 (with W in place of V). It is to this $\mathscr{F}$ that we apply the ideas of [19].

The random matching of [19], imported now to $\mathscr{F}$, is produced as follows. Fix ϵ, s and set

$$D_i = e^{-\epsilon i(k-1)} D \qquad \text{for } i = 0, \ldots, s\,,$$

$$\beta = \epsilon\, e^{-\epsilon k} \frac{1 - e^{-\epsilon s}}{1 - e^{-\epsilon}}$$

(so $\beta \approx 1$ for ϵ small, ϵs large). Set $\mathscr{F}_0 = \mathscr{F}$, $V_0 = V$, and repeat the following procedure for $i = 1, \ldots, s$.

Let X_i be a random subset of $\mathscr{F}_{i-1}$ chosen according to

$$\Pr(A \in X_i) = \epsilon / D_{i-1}$$

for each $A \in \mathscr{F}_{i-1}$, these events mutually independent. Let

$$Y_i = \{A \in X_i : A \neq B \in X_i \Rightarrow A \cap B = \emptyset\} ,$$

$$V_i = V_{i-1} \setminus \cup_{A \in X_i} A ,$$

$$\mathcal{F}_i = \{A \in \mathcal{F} : A \subseteq V_i\} .$$

Finally, the desired random matching is $M = Y_1 \cup \cdots \cup Y_s$.

As shown in Proposition 3.1 of [19], not only does M have (expected) size close to n/k, but the following more precise properties hold for $1 \leq i \leq s$ (again, as $D \to \infty$).

(i) For $1 \leq l \leq 2k-1$ and $v_1, \ldots, v_j \in V$,

$$\Pr(v_1, \ldots, v_j \in V_i) \sim e^{-\epsilon il} ;$$

(ii) For $A \in \mathcal{F}$,

$$\Pr(A \in Y_i) \sim (\epsilon/D) e^{-\epsilon(k+i-1)} ;$$

(iii) For $v \in V$, $d_{\mathcal{F}_i}(v)/D_i$, conditioned on $\{v \in V_i\}$, converges in probability to 1.

[Our interest here is really in (ii), but its proof—by induction on i—requires simultaneously proving (i) and (iii), and we include them to give a better idea of what is going on.]

In particular (ii) implies, for each $A \in \mathcal{F}$,

$$\Pr(A \in M) \sim \beta/D \sim 1/D \qquad \text{as } \epsilon \to 0 , \quad \epsilon s \to \infty . \tag{7}$$

[More precisely, we have

$$\lim_{\epsilon \to 0, \epsilon s \to \infty} \lim_{D \to \infty} D \Pr(A \in M) = 1 .$$

Limits below are also to be understood in this way.]

For Theorem 1.5 we need, in addition to (ii), approximate independence of the events $\{A \in M\}$, $\{B \in M\}$ for disjoint A, B:

$$\Pr(A, B \in M) \sim (\beta/D)^2 \sim D^{-2} \qquad \text{as } \epsilon \to 0 , \quad \epsilon s \to \infty . \tag{8}$$

This requires some extension of (i)–(iii): We should change $2k-1$ to $3k-1$ in (i) (this does not affect the proof) and add the following analogues for $1 \leq i \leq j \leq s$.

(i′) For $1 \leq l \leq 2k-1$, $A \in \mathcal{F}$ and $v_1, \ldots, v_l \in V \setminus A$,

$$\Pr(A \in Y_i, v_1, \ldots, v_l \in V_j) \sim (\epsilon/D) e^{-\epsilon(k+i-1+jl)} ;$$

(ii′) For A, $B \in \mathcal{F}$ with $A \cap B = \emptyset$,

$$\Pr(A \in Y_i, B \in Y_j) \sim (\epsilon/D)^2 e^{-\epsilon(2k+i+j-2)} ;$$

(iii′) For $A \in \mathcal{H}$ and $v \in V \setminus A$, $d_{\mathcal{F}_j}(v)/D_j$, conditioned on $\{A \in Y_i,\ v \in V_j\}$, converges in probability to 1.

Of course, (8) is implied by (ii′).

Proofs of (i′)–(iii′) are straightforward adaptations of the proofs of (i)–(iii) given in [19], to which we refer for details as it does not seem desirable to repeat those fairly lengthy arguments here.

Now recall that, for each $A \in \mathscr{H}$, $\mathscr{F}$ contains $Dt(A)$ edges that are extensions of copies of A. We may thus consider the random matching N of $\mathscr{H}$ consisting of those edges having extensions in M, and from (7) and (8) conclude that

$$p_A := \Pr(A \in N) \sim t(A)\,, \tag{9}$$

for each $A \in \mathscr{H}$, and for disjoint $A, B \in \mathscr{H}$

$$p_{AB} := \Pr(A, B \in N) \sim t(A)t(B)\,. \tag{10}$$

Notice that (9) already gives Theorem 1.2, since it implies $E[|N|] \sim t(\mathscr{H})$. For Theorem 1.5, let $c : \mathscr{H} \rightarrow \mathbf{R}^+$ [write c_A for $c(A)$] and set $\xi = c(N) = \Sigma_{A \in N}\, c_A$. Then $\mu_\xi := E[\xi] = \Sigma\, p_A c_A \sim \Sigma\, t(A)c_A$ and by (9) and (10)

$$\begin{aligned}\mathrm{Var}[\xi] &= \sum \sum c_A c_B (p_{AB} - p_A p_B) \\ &= \sum c_A^2 p_A + o(\mu_\xi^2)\,.\end{aligned}$$

Thus Chebyshev's inequality gives

$$\xi \mu_\xi^{-1} \xrightarrow{P} 1$$

provided that $\Sigma\, c_A^2 p_A = o(\mu_\xi^2)$. In particular, this is true for each $c \in \{c_1, \ldots, c_l\}$, and the theorem follows.

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